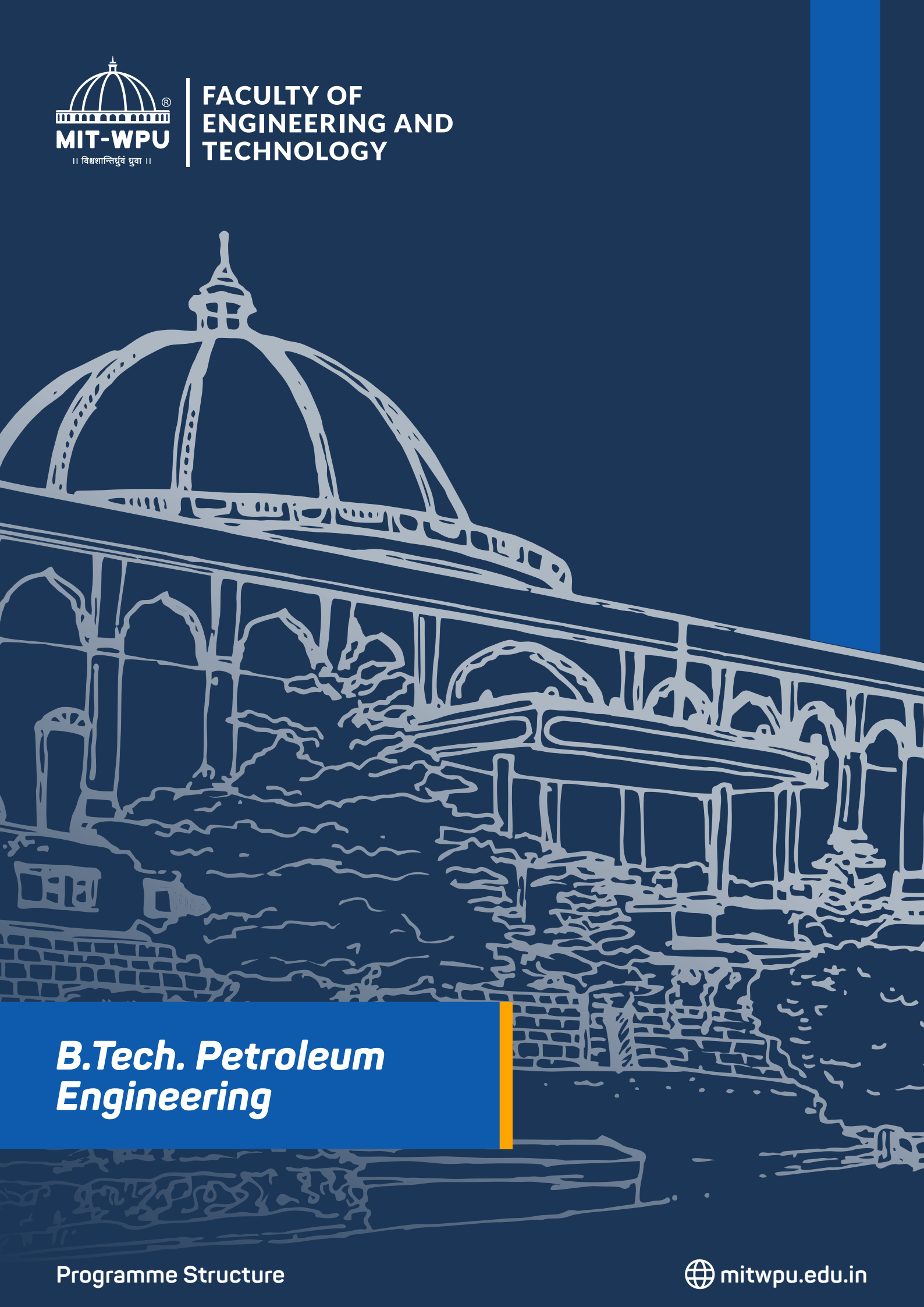




B.Tech. Petroleum Engineering

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	Department of Petroleum Engineering
Programme Name	B.Tech. Petroleum Engineering

Semester	Odd (I)	Even (II)	Total Credits
First Year	21	21	42
Second Year	22	22	44
Third Year	21	22	43
Fourth Year	17	19	36

+++++ COURSE BASKET +++++

Course Type	Description
Programme Core	Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU
Programme Electives	Open electives under the programme allow students to specialise in a particular area connected to their major.
University Core	Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students.
University Electives	Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core.

Semester I

Semester	Course Type	Course Name / Course Title	Total Credits
I	UC	Indian Constitution	1
I	UC	Environment and Sustainability	1
I	UC	Yoga - I	1
I	UC	Social Leadership Development Program	1
I	UC	Financial Literacy	1
I	PF	Linear Algebra and Differential Calculus	3
I	PF	Engineering Physics	3
I	PF	Engineering Graphics	3
I	PF	Engineering Chemistry	3
I	PF	Ideas and Innovations in Manufacturing	1
I	PF	Integral Calculus	3
		Total Credits:	21

Semester II

Semester	Course Type	Course Name / Course Title	Total Credits
II	UC	Yoga - II	1
II	UC	Co-creation	1
II	UC	AI for everyone	2
II	UC	Foundation of Peace	2
II	UC	Indian Knowledge System (General)	2
II	UC	Sports	1
II	PF	Engineering Mechanics	3
II	PF	Programming and Problem Solving (Linux based)	3
II	PF	Basics of Electrical and Electronics Engineering	3
II	PM	Petroleum Engineering Practices	3
		Total Credits:	21

Semester - III

Semester	Course Type	Course Name / Course Title	Total Credits
III	UC	Spiritual and Cultural heritage: Indian Experience	2
III	UC	Research Innovation Design Entrepreneurship (RIDE)	1
III	UE	University Electives - I	3
III	PM	Differential Equations and Transform Techniques	3
III	PM	Fluid Mechanics for Petroleum Engineers	3
III	PF	Petroleum Field Instrumentation and Control	3
III	PM	Petroleum Geology-I	3
III	PF	Reservoir Engineering - I	3
III	PF	Unit operations in Petroleum Industry	1
		Total Credits:	22

SEMESTER - IV

Semester	Course Type	Course Name / Course Title	Total Credits
IV	UC	Rural Immersion	1
IV	UC	Life Transformation Skills	1
IV	UE	University Electives - II	3
IV	PM	Drilling Operations	4
IV	PM	Statistical Methods in Oil & Gas	3
IV	PM	Petroleum Geology-II	3
IV	PM	Petroleum Exploration	3
IV	PM	Heat Transfer in Well bores	2
IV	PF	Indian Knowledge System (Sci.& Tech.)	2
		Total Credits:	22

Semester - V

Semester	Course Type	Course Name / Course Title	Total Credits
V	UC	Managing Conflicts Peacefully: Tools and Techniques	2
V	UE	University Electives - III	3
V	PE	Program Elective - I	4
V	PM	Petroleum Production Engineering-I	2
V	PF	Numerical Methods	3
V	PM	Reservoir Engineering - II	3
V	PM	Secondary Recovery	1
V	PM	Formation Evaluation	3
		Total Credits:	21

Semester - VI

Semester	Course Type	Course Name / Course Title	Total Credits
VI	University Core	National Academic Immersion Program	2
VI	Program Electives	Program Elective - II	4
VI	Program Foundation	Geomechanics	1
VI	Program Major	Well Engineering and Design	3
VI	Program Major	Petroleum Production Engineering - II	3
VI	Program Foundation	AI in Oil and Gas	3
VI	Program Major	Enhanced Oil Recovery	3
VI	Program Capstone Project/Seminar and Internships	Seminar	3
		Total Credits:	22

Semester - VII

Semester	Course Type	Course Name / Course Title	Total Credits
VII	Program Electives	Program Elective - III	4
VII	Program Capstone Project/Seminar and Internships	Internship	13
		Total Credits:	17

Semester - VIII

Semester	Course Type	Course Name / Course Title	Total Credits
VIII	PE	Program Elective - IV	4
VIII	PM	Petroleum Production Engineering - III	3
VIII	Program Capstone Project/Seminar and Internships	Capstone Project	12
		Total Credits:	19

Program Electives

Semester	Course Type	Course Name / Course Title	Total Credits
V	PE - I	Unconventional Hydrocarbon Resources	4
V	PE - I	AI for downstream industry	4
V	PE - I	Well Cementing Practices	4
V	PE - I	Pipeline Design & Engineering	4
V	PE - I	Well Completions and Services	4
VI	PE - II	Petroleum Economics	4
VI	PE - II	Machine Learning in Petroleum Engineering	4
VI	PE - II	Advanced Drilling Engineering	4
VI	PE - II	Natural Gas Engineering	4
VI	PE - II	Multi-phase Flow: Theory and Applications	4
VII	PE - III	Applied Geophysics	4
VII	PE - III	Application of ML in Petroleum Industry	4
VII	PE - III	Well Control	4
VII	PE - III	Reservoir Simulation and Modelling	4
VII	PE - III	Offshore Production Engineering	4
VIII	PE - IV	Risk Analysis and Petroleum Exploration	4
VIII	PE - IV	Optimization Methods in Petroleum Engineering	4
VIII	PE - IV	Deepwater Drilling	4
VIII	PE - IV	Reservoir Characterization for Emerging Technologies	4
VIII	PE - IV	Production Enhancement and Optimization	4

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.